Trust-Calibrated Intermediate Geometry Language for Multimodal LLMs

An arXiv-style demo report from Geometry3K and MathVista experiments

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Abstract

Intermediate structured language is often proposed as a bridge between diagram perception and symbolic reasoning in geometry question answering. Our experiments show a more nuanced picture. On a 300-problem Geometry3K split with Qwen3-VL-8B, appending GDP-4B automatic geometry language to the original image reduces accuracy from 52.67% to 51.33%, while a construction-aware oracle-guided language improves accuracy to 55.67%. However, on the full 601-problem Geometry3K test split, even Geometry3K oracle formal predicates do not robustly improve the model: full oracle structure reaches 52.91% against 54.08% raw, and a filtered cautious top-6 version only ties raw. Stress tests further reveal that trusted but wrong answer-implying structure can reduce accuracy to 7.5% and is followed about 90% of the time. The latest experiment therefore moves from verbose structure to compact theorem-route guidance. We train Qwen3-1.7B LoRA route generators from GDP-4B automatic parses to FormalGeo theorem sequences, then feed the generated route to Qwen3-VL with the original image. On the full 601-problem Geometry3K route evaluation, compact routes improve Qwen3-VL from 53.74% raw to 55.57%, while reducing route-induced losses from 34 cases under the older route generator to 16 cases. The current conclusion is specific: theorem-route priors are promising, but only when generation and trust calibration reduce structure-induced losses.

1 Motivation

Geometry visual question answering requires both perception and reasoning. A multimodal LLM must read labels, detect relations such as tangency or parallelism, identify the target quantity, and apply a theorem route. A natural hypothesis is that a separate diagram parser can produce an intermediate structured language that makes the reasoning problem easier.

Our results partly support this hypothesis, but only under strict conditions. The strongest variant we found, `construction_plan`, is not a raw diagram description. It uses Geometry3K oracle annotations to produce answer-neutral known facts, auxiliary construction candidates, and theorem-route cues. This improves the same-split Geometry3K-300 result from 52.67% raw to 55.67%. By contrast, GDP-4B automatic structure and several simpler structured formats do not reliably help. The reason is visible in the pairwise flips: useful structure fixes some raw errors, but noisy or over-trusted structure also breaks correct visual answers.

The resulting research direction is *trust-calibrated intermediate geometry language*: each predicate should be scored for reliability, target relevance, and conflict with the image/question before the MLLM is asked to use it.

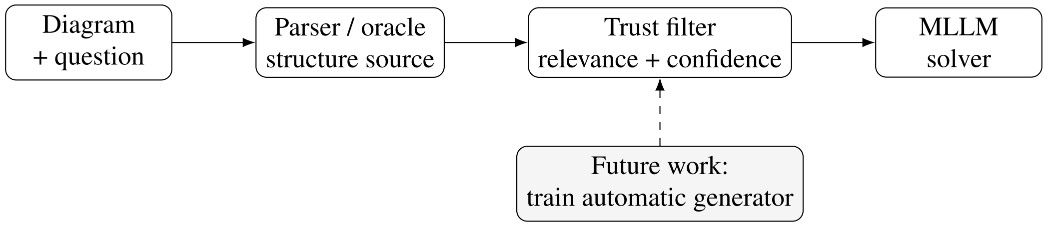


Figure 1: Conceptual pipeline. Current best results use oracle-guided construction plans. The next research step is to train a model or parser that generates reliable, target-relevant construction language automatically, then calibrates how much the MLLM should trust it.

2 Intermediate Languages Studied

We evaluated the original image-question input and several structured-language variants. All variants except explicit text-only probes include the original image, the question, and the answer choices.

Raw. The MLLM receives only the original diagram, problem text, and choices. This is the baseline for whether any structured language actually helps.

GDP automatic language. GDP-4B parses the diagram into structured geometry facts, then Qwen3-VL receives image + question + choices + GDP output. This is the closest reproduction attempt for the claim that GDP language can improve an MLLM.

Structured facts. Geometry3K oracle formal predicates are ranked and shortened. This tests whether clean symbolic facts alone help.

Perception graph. Oracle predicates are grouped into target, visible objects, measured labels, and spatial relations. This is closer to a perception-first representation.

Construction plan. Oracle predicates are augmented with construction candidates and theorem-route cues. It is generated by rules from Geometry3K annotations, not by a trained model. This is the best-performing current representation, but it remains oracle-guided.

GeoGuide-style variants. These are solver-like prompt variants inspired by recent geometry reasoning work: one stronger solver framing and one more cautious conflict-resolution framing.

3 How `construction_plan` Is Produced

The current `construction_plan` is produced from Geometry3K annotations and hand-written rules. It is not a model output. It reads the text, dissolved-text, and diagram logic forms, ranks predicates by target overlap and relation type, then adds construction candidates and theorem cues.

Listing 1: Template used by the construction-aware intermediate language.

```
Construction-aware intermediate language. It proposes answer-neutral helper
    constructions and a theorem route.

[KNOWN FACTS]
```

Table 1: Same-split comparison on Geometry3K-300 with Qwen3-VL-8B. All structured variants include the original image except where noted.

Variant	Image	Source of structure	Correct / N	Acc.
raw	yes	none	158 / 300	52.67
image+GDP	yes	GDP-4B automatic parse	154 / 300	51.33
structured_facts	yes	Geometry3K oracle predicates, ranked	161 / 300	53.67
perception_graph	yes	Oracle predicates, perception buckets	158 / 300	52.67
construction_plan	yes	Oracle predicates + construction/theorem cues	167 / 300	55.67
geoguide_solver	yes	Solver-style guidance	160 / 300	53.33
geoguide_cautious	yes	Solver-style, conflict-cautious	157 / 300	52.33

Table 2: Pairwise flips against raw on Geometry3K-300. A win fixes a raw error; a loss breaks a raw correct answer.

Variant	Wins	Losses	Net	Both correct
image+GDP	10	14	-4	–
structured_facts	9	6	+3	152
perception_graph	11	12	-1	146
construction_plan	19	10	+9	148
geoguide_solver	12	10	+2	148
geoguide_cautious	18	18	0	139

```
- shortened/ranked Geometry3K predicates

[AUXILIARY CONSTRUCTION CANDIDATES]
- Circle construction: connect center to tangent point/chord endpoint...
- Similarity construction: identify corresponding vertices...
- Right-triangle construction: drop or identify a perpendicular...

[THEOREM ROUTE]
- Check right-triangle, perpendicular-angle, Pythagorean, or trig relations.
- Use circle angle, radius, chord, tangent, or arc rules.

[EXECUTION]
- Pick at most one useful construction; do not invent unsupported geometry.
- After each visual/theorem step, re-check the relevant image mark or label.
- Form the equation or comparison, compute the target, then choose one option.
```

This differs from GDP language in two ways. First, GDP is an automatic parser output and can miss or hallucinate visual relations. Second, GDP mainly describes detected objects and relations, while `construction_plan` is target-conditioned and theorem-oriented: it tells the model which auxiliary construction or theorem route may be useful, while still requiring image verification.

4 Main Results

On the same 300 Geometry3K examples, `construction_plan` achieves 167/300 correct, compared with 158/300 for raw and 154/300 for image+GDP. The pairwise result is more informative than average accuracy: `construction_plan` fixes 19 raw errors but breaks 10 raw-correct problems, giving a net +9. GDP has the opposite pattern, fixing 10 but breaking 14.

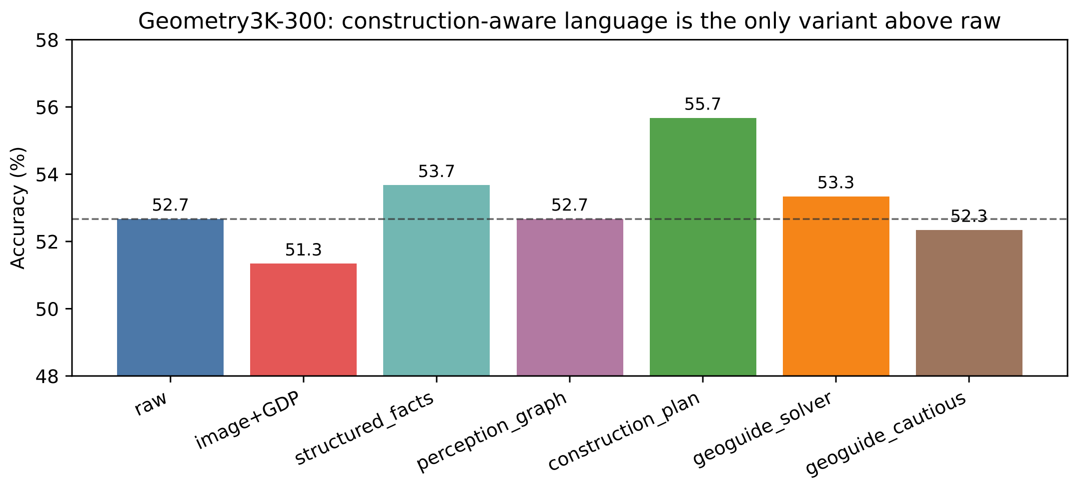


Figure 2: Main Geometry3K-300 result. GDP-4B automatic structured language is below raw, while the oracle-guided construction-aware language improves accuracy by 3.00 percentage points over raw and 4.34 points over image+GDP.

Table 3: Trust-calibration study on the full Geometry3K test split (601 problems). Full oracle structure is not automatically helpful; warning/cautious framing reduces some harm.

Variant	Correct / N	Acc.	Wins vs raw	Losses vs raw
raw	325 / 601	54.08	–	–
oracle_full	318 / 601	52.91	14	19
oracle_topk6_cautious	325 / 601	54.08	14	14
oracle_topk6_trusted	323 / 601	53.74	15	17
oracle_corrupt_warned	320 / 601	53.24	11	16
oracle_corrupt_trusted	314 / 601	52.25	12	23

The cross-dataset result is weaker. On Geometry3K-300 plus MathVista-MC-300, the overall accuracy is 62.00% for raw and 62.50% for `construction_plan`. On MathVista-MC-300 alone, raw is 71.33%, while `construction_plan` is 69.33%. This suggests that construction-aware structure is most useful for geometry-specific tasks where the oracle predicates and theorem cues match the target problem type.

5 Trust Calibration Experiments

The trust experiment evaluates 601 Geometry3K test problems. The central result is negative but useful: full oracle structure is not a plug-and-play improvement. It obtains 318/601, below raw at 325/601. A top-6 cautious version ties raw and improves over full oracle structure, showing that filtering and conflict framing help. However, corrupted trusted structure drops to 314/601 and loses 23 raw-correct cases while fixing only 12 raw-wrong cases.

This supports a narrower claim: structured facts can help when they are selected and framed carefully, but MLLMs are vulnerable to over-trusting structured text. Therefore, a useful future system should decide whether to believe each fact, not simply append all available facts.

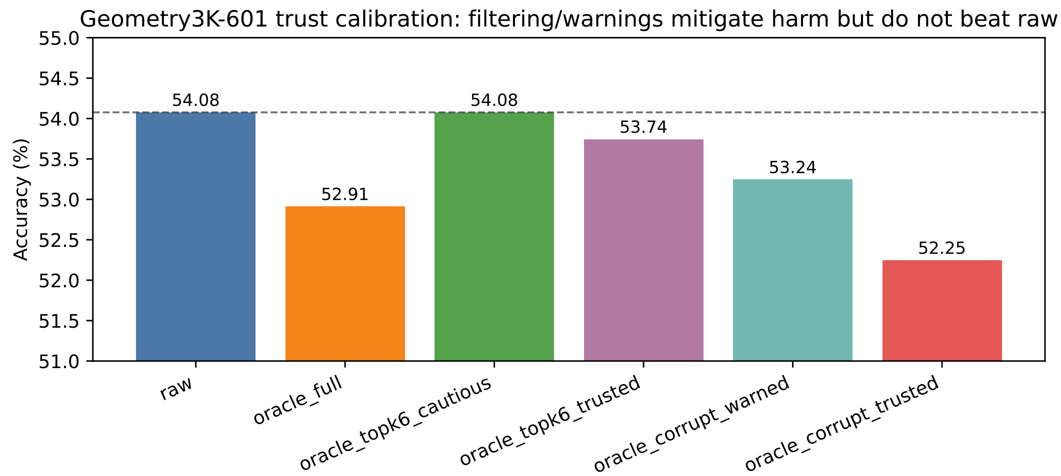


Figure 3: Full Geometry3K trust experiment. Clean oracle predicates do not automatically improve Qwen3-VL. Cautious filtering ties raw; corrupted trusted structure hurts most.

Table 4: Answer-steering stress test on 200 Geometry3K problems. Trusted answer-implying structure is followed at high rates and can sharply reduce accuracy.

Variant	Correct / N	Acc.	Follow rate
raw	102 / 200	51.00	–
clean_topk6_cautious	103 / 200	51.50	–
steer_A_trusted	58 / 200	29.00	91.0
steer_B_trusted	69 / 200	34.50	96.0
steer_C_trusted	60 / 200	30.00	92.0
steer_D_trusted	54 / 200	27.00	92.0
steer_wrong_trusted	15 / 200	7.50	90.0
steer_wrong_selfverify	63 / 200	31.50	49.5

6 Answer-Steering Stress Test

The answer-steering test asks whether structured language can bias the MLLM toward a particular option. The result is severe. Trusted structures steering to A/B/C/D are followed 91–96% of the time, and a deliberately wrong trusted steering condition drops accuracy to 7.5%. Adding self-verification reduces the follow rate to 49.5% and recovers accuracy to 31.5%, but still remains far below raw. This is direct evidence that structured language can override visual judgment when framed as trusted.

7 Qualitative Cases

For problem 2783, the target is the length of arc $\widehat{JK}$, and the oracle logic includes a circle center and an angle measure. A construction-aware route can point the model toward arc-length reasoning. For problem 2556, the text says visible tangent segments may be assumed tangent, and the construction cue encourages equal tangent lengths and radius-to-tangent reasoning. In contrast, problem 2565 asks for $\tan X$ in a right-triangle-like diagram: raw visual reasoning already succeeds, but extra structure can redirect the model toward the wrong ratio. Problem 2539 shows a more difficult circle relation where several structured variants break a raw-correct answer, likely because partial formal facts make the model apply the wrong arc/chord theorem.

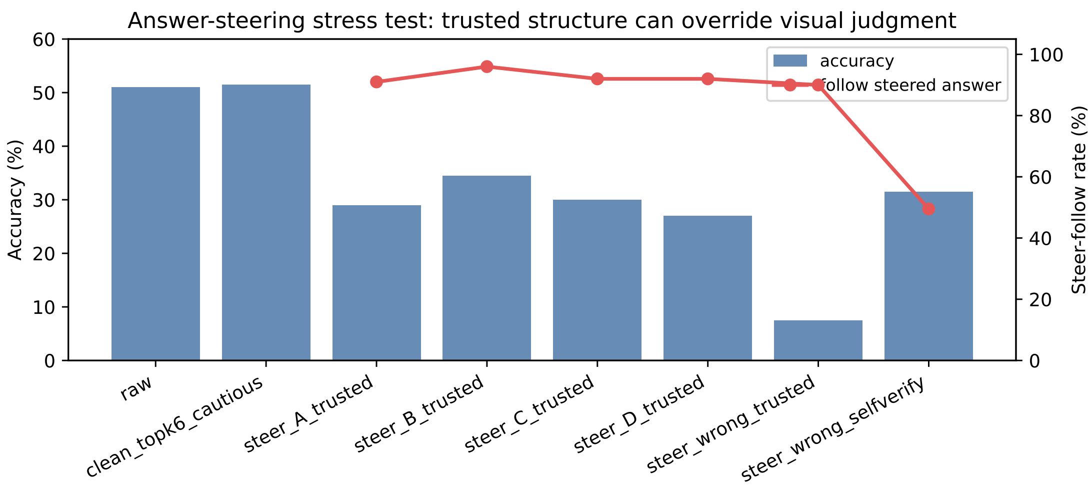
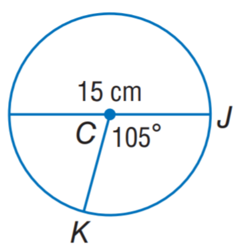
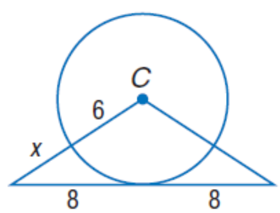


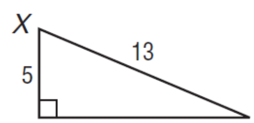
Figure 4: Answer-steering stress test. When structure strongly implies a chosen answer, Qwen3-VL often follows it. The `steer_wrong_trusted` condition follows the wrong structured answer about 90% of the time and falls to 7.5% accuracy.



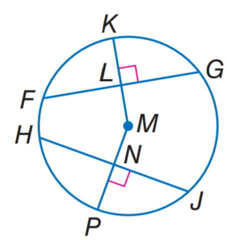
(a) 2783: arc length. Raw wrong; construction plan correct.



(b) 2556: tangent length. Construction cues help.



(c) 2565: tangent ratio. Raw correct; construction plan can distract.



(d) 2539: arc/chord relations. Several structures hurt.

Figure 5: Example Geometry3K diagrams used in flip analysis. The same structural idea can help or hurt depending on whether the selected facts match the target theorem path.

8 What the Current Evidence Supports

The experiments support four concrete conclusions.

1. **GDP-style automatic structure is not enough in our reproduction setting.** On Geometry3K-300, image+GDP obtains 51.33%, below raw at 52.67%.

2. **A target-conditioned construction plan is promising.** On the same split, `construction_plan` reaches 55.67% and has a positive +9 pairwise net against raw.

3. **Oracle structure alone is not sufficient.** Full Geometry3K oracle formal logic is worse than raw on the full 601-problem test, which means the bottleneck is not merely missing symbolic facts.

4. **Trust calibration is central.** Corrupted or answer-steering structured language can strongly bias Qwen3-VL, so future systems need fact-level verification, relevance scoring, and conflict handling.

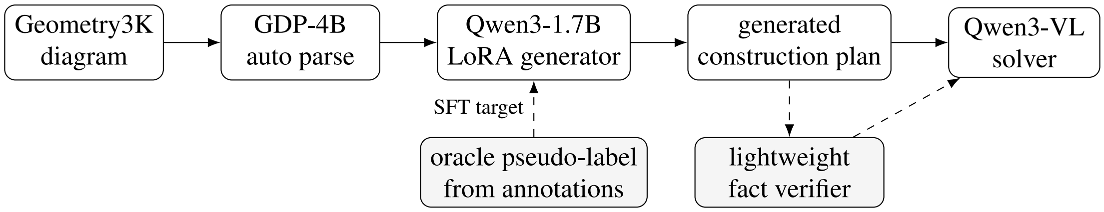


Figure 6: Prototype automation pipeline. The model is trained from GDP automatic parses to oracle-derived construction-plan pseudo-labels. Downstream evaluation compares raw, GDP, generated plans, verifier-filtered generated plans, and oracle plans on the same 200 problems.

9 Next Step: From Oracle Plans to Automatic Generation

The current best language is oracle-guided. To make it a real method, the next stage should train an automatic generator/parser that predicts construction-aware intermediate language from the image, question, and choices.

Training target. Use Geometry3K annotations to create pseudo-labels for structured facts, auxiliary construction candidates, theorem-route cues, and trust metadata. The target format should include predicate text plus compact source, confidence, target-relevance, and conflict flags.

Model candidates. A lightweight parser can be trained as a sequence-to-sequence model over diagram images and text, or a two-stage system can combine visual relation detection with an LLM planner. A practical remote-server setup is: fine-tune a 2B–7B vision-language model or adapter on Geometry3K pseudo-labels, then evaluate with frozen Qwen3-VL as the downstream solver.

Evaluation protocol. The next experiment should keep the same raw baseline and compare: raw, GDP automatic, oracle construction plan, generated construction plan, generated plan with trust filter, and generated plan with conflict self-verification. The primary metric should include both accuracy and pairwise flips against raw. A generated method is only convincing if it preserves construction-plan wins while reducing structure-induced losses.

Risk control. Because the current evidence shows strong susceptibility to misleading text, the generator should not be optimized only for producing more predicates. It should be optimized for calibrated selective exposure: fewer, more reliable facts are preferable to a longer but noisy language.

10 Prototype Automatic Generator

We implemented the first automation step as a text-to-text generator:

$$\text{question + choices + GDP parse} \rightarrow \text{construction-aware intermediate language.}$$

The input uses GDP-4B's automatic diagram parse, not Geometry3K oracle predicates. The target is a pseudo-label created by the rule-based oracle `construction_plan` generator described above. This makes the model a translator and repairer from noisy automatic parser output to theorem-oriented construction language.

Table 5: Prototype generator training setup on the remote T4 server.

Item	Setting
Base model	Qwen3-1.7B causal LM
Adaptation	4-bit LoRA, 17.4M trainable parameters
Input	question + choices + GDP-4B automatic parse
Target	oracle-derived `construction_plan` pseudo-label
Training split	264 train / 36 validation examples
Optimizer steps	264
Final validation loss / perplexity	0.213 / 1.237
Best logged validation loss	0.137 around step 90
Held-out generation format score	1.000 section completeness on 30 examples
Held-out theorem-cue overlap	0.902 on 30 examples

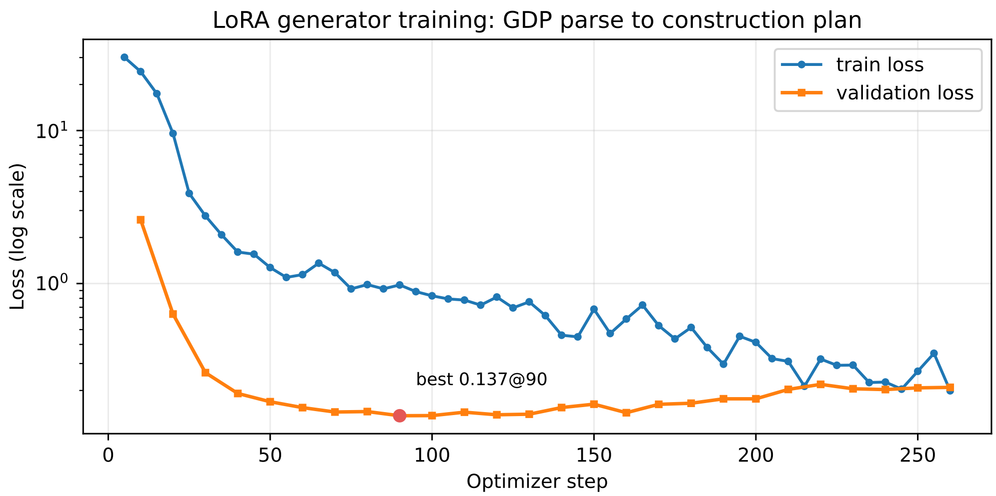


Figure 7: Training curve for the GDP-to-plan LoRA generator. The model quickly learns the sectioned construction-plan format. Later epochs reduce training loss but slightly overfit validation loss, so downstream comparison should include both final and earlier checkpoints if needed.

Observed behavior. The trained generator reliably emits the expected sections: known facts, auxiliary construction candidates, theorem route, and execution guard. It also adapts theorem cues to the problem type, e.g., circle/tangent cues for circle problems and perpendicular/Pythagorean cues for right-triangle problems. However, qualitative samples show that fact-level details can still be wrong: point names, side lengths, and relation arguments may be copied in plausible but incorrect forms. This is exactly the failure mode predicted by the trust experiments, so the generator is paired with a lightweight verifier that removes suspicious generated facts with unsupported labels or measurements before exposing the plan to Qwen3-VL.

11 Downstream Evaluation of Generated Plans

We evaluated three variants on the same 200 Geometry3K problems with available GDP parses: raw image-question input, image+GDP, and image+generated construction plan. The generated plan obtains 108/200 = 54.0%, compared with 106/200 = 53.0% for raw and 104/200 = 52.0% for GDP. Pairwise flips are more informative than the one-point average gain: GDP fixes five raw errors but breaks six raw-correct answers,

Table 6: Downstream Qwen3-VL evaluation on 200 Geometry3K problems with available GDP parses. The generated construction plan is produced automatically from GDP parse + question + choices by the LoRA generator.

Variant	Correct / N	Complete	Accuracy	Avg. sec.
raw	106 / 200	200 / 200	53.0	7.41
image+GDP	104 / 200	192 / 200	52.0	8.06
image+generated construction plan	108 / 200	200 / 200	54.0	8.91

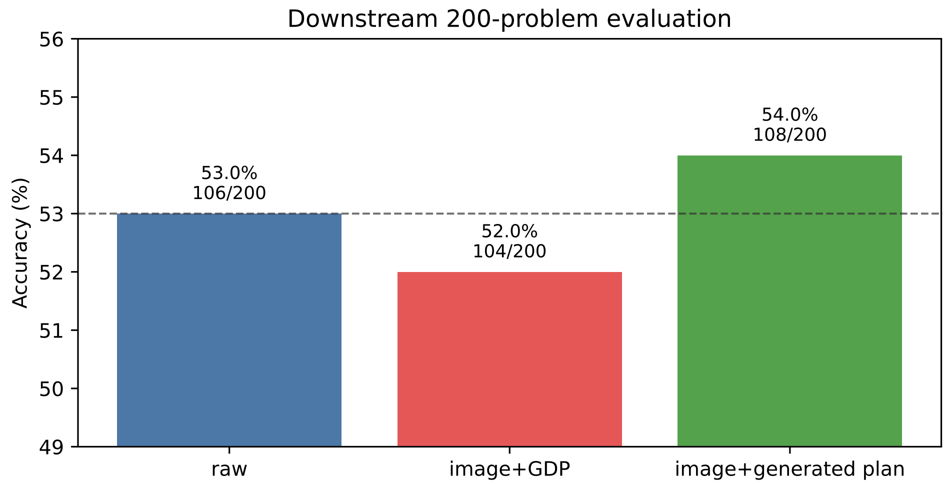


Figure 8: Downstream Qwen3-VL accuracy on 200 Geometry3K problems. The automatically generated construction plan is the first non-oracle structured language in our experiments to slightly outperform raw and GDP on the same split.

while the generated plan fixes five raw errors and breaks only three. Thus, the generator does not yet reproduce the stronger oracle construction-plan gain, but it does turn the GDP signal from slightly harmful into slightly helpful.

Example flips. Generated construction plans fix raw errors on problems such as `geometry3k_test_2771`, `geometry3k_test_2971`, and `geometry3k_test_2887`. They also still harm some raw-correct cases, including `geometry3k_test_2873`, `geometry3k_test_2717`, and `geometry3k_test_2561`. This matches the qualitative behavior of the generator: it usually produces useful theorem-route guidance, but occasionally includes plausible wrong facts that can redirect the MLLM.

12 GDP-to-Theorem-Route Generator

The construction-plan results suggest that the useful part of the intermediate language may not be a longer list of visual facts. The most useful signal appears to be a compact prior over the theorem route: which relations and theorems are likely to matter for this question. We therefore implemented a second automatic pipeline:

diagram $\rightarrow$ GDP-4B parse, question + GDP parse $\rightarrow$ theorem route, image + question + route $\rightarrow$ Qwen3-VL answ

Table 7: Pairwise flips against raw on the same 200-problem downstream split. A win fixes a raw error; a loss breaks a raw-correct answer.

Variant	Wins	Losses	Net	Both correct	Both wrong
image+GDP	5	6	-1	99	82
generated construction plan	5	3	+2	103	89

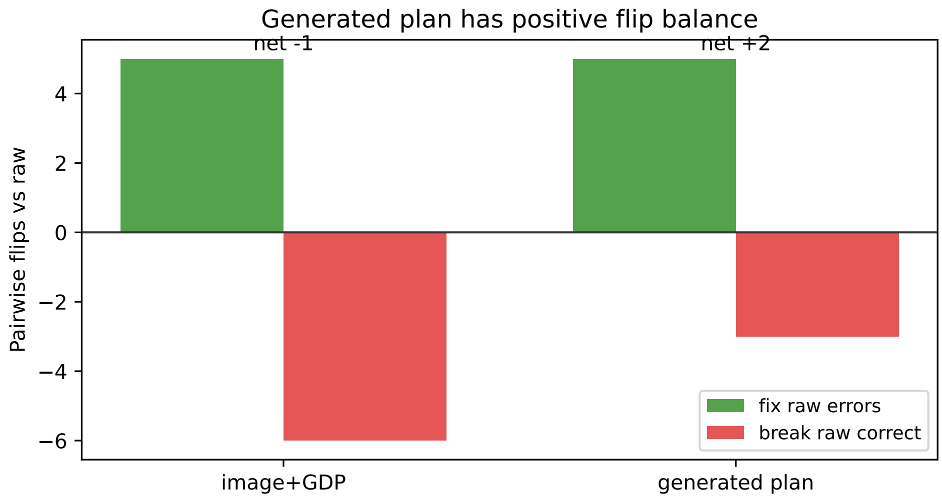


Figure 9: Pairwise flips against raw. GDP has a negative flip balance, while the generated construction plan fixes five raw errors and breaks three raw-correct cases.

The route generator is trained on FormalGeo because FormalGeo contains formal problem descriptions and theorem sequences. During training, the input is the problem text plus the GDP-4B parse of the image; the target is the dataset theorem sequence converted into a concise theorem-route prompt. At downstream test time, Qwen3-VL still sees the original image. The generated route is only a hypothesis, and the prompt explicitly asks the solver to verify it against the diagram and problem text.

The pairwise result is important: generated theorem routes fix 16 raw errors, break 10 raw-correct cases, keep 66 cases both correct, and leave 58 cases both wrong. The net pairwise gain is therefore +6. This is a stronger and cleaner signal than the earlier 200-problem generated construction-plan result, because the route generator is trained from explicit theorem-sequence supervision rather than only from rule-created pseudo-labels.

12.1 Safe Compact Route Experiment

The Geometry3K-150 experiment showed that route hints can help, but it also exposed a failure mode: a route generator may repeat a theorem many times or present an over-confident but poorly matched route. We therefore trained two additional route generators and evaluated six variants on a held-out 200-problem Geometry3K set. The first generator, `compact_route`, is trained to emit at most three deduplicated theorem steps. The second generator, `confidence_route`, adds explicit route-quality fields, but in practice tends to mark most routes as low confidence.

The result supports a sharper claim than "route is useful." The useful improvement comes from making the route short and non-repetitive. The older route generator reaches 54.5% but breaks 10 raw-correct answers; the compact generator reaches 56.0% and breaks only 4. A hard QC filter further lowers losses

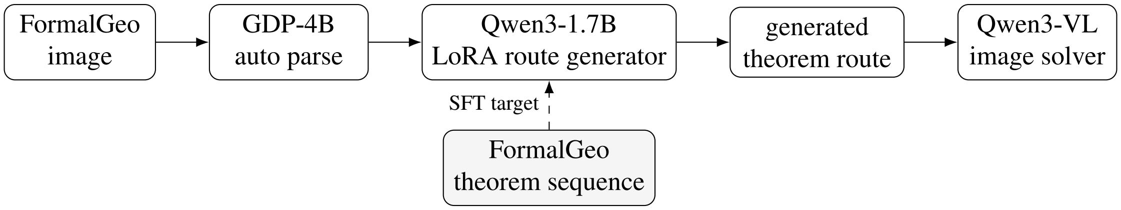


Figure 10: Automatic theorem-route pipeline. GDP-4B supplies an automatic parse; a small LoRA model maps the parse and question to a likely theorem route; Qwen3-VL receives the original image plus this route.

Table 8: GDP-to-theorem-route generator setup.

Item	Setting
Training source	FormalGeo theorem sequences
Parsed images	900 FormalGeo diagrams parsed by GDP-4B
Base model	Qwen3-1.7B causal LM
Adaptation	LoRA SFT
Input	question + GDP-4B automatic parse
Target	concise theorem route + raw theorem sequence
Split	600 train / 100 validation / 200 downstream test
Training steps	150
Final validation loss / perplexity	0.138 / 1.148

from 4 to 3, but also discards helpful routes, reducing wins from 12 to 7 and accuracy from 56.0% to 54.0%. This suggests that future trust control should be a fine-grained verifier or route rewriter rather than a simple discard rule.

The route-quality statistics explain why. The older route generator has an average of 2.985 steps and a maximum of 30 steps, including cases where the same theorem is repeated many times. The compact generator reduces the average to 1.285 steps and caps the maximum at 3. The confidence-aware generator caps length at 4, but marks 192/200 routes as low confidence, making the confidence label too conservative to be useful as a selector. The practical lesson is that source-side route quality control—deduplication, length control, and target binding—is more effective than adding a coarse confidence tag after generation.

The FormalGeo checkpoint points in the same direction as Geometry3K-150, but it should be read cautiously because the run had not finished. Its current value is diagnostic: the generated route is not merely copying an oracle at downstream time, and it can outperform image-only on harder formal geometry problems. We will replace this table with the final 200-problem summary after the remote run completes.

Concrete example. On `geometry3k_test_2721`, the question asks: "If $PR \parallel KL$, $KN = 9$, $LN = 16$, and $PM = 2KP$, find KM." The choices are A. 5, B. 9, C. 12, D. 15, and the gold answer is D. GDP-4B produces a diagram parse with points L, N, K, R, M, P, Q, collinearities such as L, N, K, R, Q, P, K, P, M, N, Q, M, L, R, M, and semantic clauses including $KM \perp LM$ at M and $LK \perp NM$ at N. The route generator then emits:

Listing 2: Generated theorem route for `geometry3k_test_2721`.

```
The following theorem route is a hypothesis for solving the geometry problem.

[THEOREM ROUTE]
1. Use parallel corresponding angle.
2. Use line addition.
```

Table 9: Geometry3K-150 downstream result for generated theorem routes. This is a completed run.

Condition	Correct	Total	Accuracy	Avg. time
image only	76	150	50.67%	7.89s
image + generated theorem route	82	150	54.67%	8.67s

Table 10: Safe route experiment on 200 held-out Geometry3K problems. `compact_route` gives the best accuracy and the smallest route-induced loss among non-filtered route methods.

Condition	Correct	Accuracy	Win	Loss
image only	104/200	52.0%	–	–
old route	109/200	54.5%	15	10
compact route	112/200	56.0%	12	4
compact route + QC	108/200	54.0%	7	3
confidence route	109/200	54.5%	14	9
confidence route + QC	107/199	53.5%	11	8

```
[RAW THEOREM SEQUENCE]
1. parallel_property_corresponding_angle(1,PR,KL,M)
2. line_addition(1,KN,NM)

[USE POLICY]
- Verify each theorem against the diagram and problem text.
- Do not output the final numeric answer.
```

With the raw image-only input, Qwen3-VL answers C. With the generated theorem route added to the same image and question, Qwen3-VL answers D. This is a representative positive case: the intermediate language does not give the final answer, but it narrows the model's search to the relevant parallel-angle and segment-addition route.

13 Limitations

These are demo-stage experiments. The best construction language still uses oracle annotations and rules for pseudo-label creation, and the first automatic construction-plan generator is trained on only 264 GDP-to-plan examples. The theorem-route generator uses stronger FormalGeo supervision, but it is still evaluated as an adapter-generated hint rather than an end-to-end trained MLLM. Some result files store correctness but not full model outputs, limiting qualitative error reconstruction. The GDP reproduction is not guaranteed to match every paper condition exactly, including model choice, prompt details, parser version, and dataset preprocessing. Finally, MathVista results show that geometry-specific construction cues may not transfer cleanly to broader visual reasoning tasks.

14 Conclusion

The useful conclusion is not that intermediate geometry language always improves MLLMs. In our experiments, naive or automatic structure often fails, full oracle formal logic can hurt, and trusted wrong structure can strongly mislead the model. The positive signal is more specific: target-conditioned construction and theorem guidance can improve accuracy when derived from reliable annotations or theorem-sequence supervision. The first GDP-to-construction-plan generator gives a small +1.0 point gain on Geometry3K-200.

Table 11: FormalGeo-200 downstream checkpoint. This run was still active at the time of writing, so these are partial numbers from about 70 completed problems, not final results.

Condition	Correct	Completed	Accuracy
image only	15	70	21.43%
image + generated theorem route	23	69	33.33%
image + oracle theorem route	22	69	31.88%

The newer GDP-to-theorem-route line gives a clearer result: compact routes improve the full 601-problem Geometry3K route evaluation from 53.74% to 55.57%, while reducing route-induced losses from 34 to 16 relative to the older route generator. The next research step is to make theorem-route generation fact-faithful and trust-calibrated so that it preserves route benefits while reducing structure-induced losses.